

Resolution Geometry

Canonical Transport: Why the Laws Select No Connection, and the Connection Declared Second-Order Data Select

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October 2026

Abstract

Rung L58 asked for a canonical connection for latent transport built from the response Jacobian, the covariance and the memory kernel. It closes in two parts.

The negative part is a theorem. Vertical shears preserve the observation law and act transitively on the horizontal complements of the observational kernel. No connection is therefore natural with respect to the law, and no amount of law data, Fisher geometry, entropy or wall specialization selects one.

The positive part is also a theorem. Once a latent covariance P is declared, the P^{-1} -orthogonal complement $H_P = \text{im}(P D F^\top)$ is the canonical connection. It is:

- the unique minimal-norm lift and the noiseless limit of the Kalman update;
- equivariant;
- coherent across rank walls: the wall horizontal space lies in the limit of the regular ones;
- curved for nonlinear laws, with latent holonomy of order the area of the observed loop;
- with a memory kernel, the path-space Wiener–Kolmogorov lift, which equals the Rauch–Tung–Striebel (RTS) smoother.

The global transport of RG is then fixed. The observable atlas closes without extra data, by regular groupoid transport together with wall specialization and correspondences. Latent transport closes exactly when second-order latent data are declared. A seven-check battery passes 7/7 on five seeds and two mutation canaries are caught.

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1 Setting

On a regular stratum, let F be the joint law, $V = \ker DF$ the observational vertical bundle, and π the submersion to the quotient. A *connection* (Ehresmann) is a complement H with $TX = H \oplus V$. A transport generated by latent dynamics descends iff it is projectable (L42); the question is which horizontal lift of observed motion is canonical.

2 The no-go

Theorem 2.1 (Gauge transitivity; TC01). *Let B_1, B_2 be two lifts with $DF B_i = I$, spanning complements H_1, H_2 . The vertical shear*

$$\varphi = \text{id} + (B_2 - B_1) DF$$

(locally, in the linearized law) satisfies $F \circ \varphi = F$, so $\varphi \in \text{Gau}$, and $\varphi B_1 = B_2$. Hence Gau acts transitively on complements. A connection invariant under Gau, which any natural module must be, exists only if $V = 0$.

Proof. $DF(B_2 - B_1) = 0$ gives $DF \circ \varphi = DF$, and $\varphi B_1 = B_1 + (B_2 - B_1)DF B_1 = B_2$. Natural modules are Gau-invariant (L13), and the stabilizer of a complement under all shears is trivial when $V \neq 0$. $\square$

On 200 random laws the law change was 0 and the complement mismatch after the shear was 0.

Corollary 2.2. *Neither the laws nor their natural modules select a connection: not the Fisher metric, not entropy (L41), not wall specialization (L16.8), and not quantum metric selection.*

3 The theorem

Hypothesis 3.1 (H-cov). A latent covariance $P \succ 0$ is declared (a prior, or the covariance of a latent dynamics). It is latent structure, not a natural module.

Theorem 3.2 (Canonical connection; TC02). *Under H-cov, $H_P = \text{im}(P DF^\top) = V^{\perp_{P^{-1}}}$ is a complement to V , and:*

1. *the lift $L_P = P DF^\top (DF P DF^\top)^{-1}$ is the unique lift of minimal P^{-1} -norm;*
2. *L_P is the noiseless limit of the Kalman gain $P DF^\top (DF P DF^\top + R)^{-1}$ [4];*
3. *it is equivariant, $H_{g P g^\top} (DF g^{-1}) = g H_P (DF)$;*
4. *different declared P give different transports on identical laws.*

Proof. (1) Minimal-norm solution of $DF x = \delta y$ in the P^{-1} inner product. (2) Let $R \rightarrow 0$. (3) Substitution. (4) H_P depends on P . $\square$

Measured on 200 random cases:

- the complement and minimal-norm properties held in every case;
- the Kalman limit matched to relative error 3×10^{-10} ;
- the equivariance error was 0;
- the median gap between transports for two declared P was 0.85.

Theorem 3.3 (Curvature and latent holonomy; TC03, TC07; [2]). *H_P is integrable iff its O’Neill tensor vanishes. Otherwise the horizontal lift of a closed observed loop of area a returns displaced along the fibre by $O(a)$. Linear laws are flat, and the holonomy depends on P .*

For $F(x) = (x_1 + \frac{1}{2}x_3^2, x_2 + 0.8x_1x_3)$ with $P = I$:

- the holonomy is 1.6×10^{-4} , 4.0×10^{-5} and 1.0×10^{-5} for loop sides 0.02, 0.01 and 0.005, which is order 1.987;
- a linear law gives 0;
- $P = \text{diag}(1, 2, \frac{1}{2})$ gives 6.0×10^{-5} ;
- the observed loops close to $< 10^{-8}$.

A clean observed loop therefore does not certify trivial latent holonomy.

Theorem 3.4 (Memory; TC04). *If the latent process has a memory kernel, the canonical lift lives on path space, $\hat{x} = P_{\text{path}} J^\top (J P_{\text{path}} J^\top)^{-1} y$, the Wiener–Kolmogorov conditional mean. It equals the RTS smoother in the noiseless limit [3].*

For a two-dimensional latent AR(1) process observed in one dimension over 40 steps:

- the RTS smoother matches the path lift to relative error 1.3×10^{-10} ;
- the memoryless instantaneous lift differs by 56%;
- a noisy smoother ($R = 1$) fails the gate (canary).

Theorem 3.5 (Coherence across walls; TC05). *Let $DF_\varepsilon \rightarrow DF_0$ with a rank drop, under H -cov with the same P on both sides. Then $H_{P,0} \subseteq \lim H_{P,\varepsilon}$; the containment defect is linear in ε . The wall transport is the P^{-1} -orthogonal projection onto $H_{P,0}$. The components in $\lim H_{P,\varepsilon} \ominus H_{P,0}$ are those that become silent at the wall.*

Proof. $\lim V_\varepsilon \subseteq V_0$ (L16.4). Taking P^{-1} -orthogonal complements reverses the inclusion, and taking complements commutes with limits in the Grassmannian. $\square$

Measured:

- the containment defect at $\varepsilon = 10^{-4}$ is $\leq 6 \times 10^{-4}$, and it decays by 100 $\times$ when ε shrinks 100 $\times$;
- arbitrary complements, with no declared metric, give a median defect of 0.69;
- a different P at the wall gives 0.55 (caught by the canary);
- two declared P give wall transports that differ by a median of 0.80.

4 Global transport

Theorem 4.1 (Global transport of RG). *1. Observable atlas (no extra data). Within strata, transport is the invertible transition groupoid (L16.5). Across walls it is specialization: maps under kernel inclusion, correspondences otherwise. Wall geometry is obtained by re-discrimination of the specialized task family, with outcomes LOCKED, DIVERGENT, VACUOUS, UNDERDETERMINED or INCONSISTENT. This closes the observable atlas.*

2. Latent transport. Without declared data, latent transport is undetermined (Theorem 2.1). With declared P and memory kernel, it is canonical (Theorems 3.2 and 3.4), coherent across walls (Theorem 3.5), and its curvature is the latent holonomy (Theorem 3.3).

5 Ladder insertion

Rung	Statement	Status	Ref.
T1	No law-natural connection: vertical shears act transitively	no-go	Thm. 2.1
T2	Canonical connection under declared covariance	hypothesis H-cov	Thm. 3.2
T3	Curvature = latent holonomy; observed loops do not certify it	forced	Thm. 3.3
T4	Memory: path-space lift = RTS smoother	imported / forced	Thm. 3.4
T5	Coherence across walls	H: cov	Thm. 3.5
T6	Global transport theorem	H: cov for latent part	Thm. 4.1

A Reproducibility and audit

`rg_canonical_transport_attack.py` passes 7/7 on seeds 20261002, 1, 42, 777 and 2026. Canaries (a noisy smoother; a different P at the wall) are both caught.

Corrections before registration:

1. The first cross-wall check inverted a nearly singular matrix.
2. Its replacement tested $\Pi_0 = \Pi_0 \circ \lim \Pi_\varepsilon$, which holds for every projector along the wall kernel, so it was vacuous; the canary exposed this.
3. TC05 now tests containment of horizontal spaces, with controls.

Audit of the earlier cross-wall battery:

- Its A1–A4 used an invertible regular law, so inclusion and factorization passed for an unrelated wall in 100/100 cases (TC06).
- “Wall relation is many-valued” and “observable relation composes” were the literals `11>1` and `True`.
- Its holonomy check thresholded random normals.
- The reported 17 checks include a C1 absent from the code, which has 16.

Its correct conclusions are retained:

- correspondences where kernel inclusion fails;
- specialize the law, then reconstruct the geometry;
- latent transport undetermined without declared data.

References

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